

Going back to Covert Attacks.

[DRAWING ON PROFESSOR'S NOTES]

The idea is that the attack is going to have an impact on y , and cancel it from the measurements. We have that $y_k = \bar{y}_k + y_k^a$, and we're basically estimating y_k^a . The attacker will use the model of the system to compute the estimate, specifically of the process P . We call this approximated model Π .

We take the attack we inject on control channel, a_u , and send it to the estimation block. From it, we get γ , which is an approximation of y_k^a , so the impact of a^u on y .

This exact value is what I subtract from y_m , that is being sent to the controller. This is for the "covert" part of the attack. The other objective is misappropriation. In ZDA, the objective was to have the state diverge, will misappropriation means that we want to control the outputs.

The idea is that C brings y close to y_{ref} , and the attacker wants to take control of y (or some variables) to another reference. We define γ^* as the change in the outputs desired by the attacker. The output from the Π block, as we said, yields γ . We then pass this γ , alongside γ^* , into a C^a module, that outputs a^u , creating a smaller loop.

For this attack we don't need disclosure resources, but we need model knowledge (specifically the model of P , we don't need to know the model of C) and disruption resources.

Impact of knowledge of initial state of the system on detection x_0

Let's consider a simple setting given by a linear system.

$$\begin{cases} x_{k+1} = Ax_k + \bar{B} \cancel{u_k} + Ba_k \\ y_k = Cx_k + \bar{D} \cancel{u_k} + Da_k \end{cases}$$

B and D are matrices telling how the attack impacts on the system, and they're not known to the operator.

Since system is linear, we put u_k to zero, continuing the analysis only analysing the effect of the attack a_k . We get:

$$\begin{cases} x_{k+1} = Ax_k + Ba_k \\ y_k = Cx_k + Da_k \end{cases}$$

The system is in a given initial state $x(0)$. We could have three possible cases:

1. Full model knowledge, and so the operator exactly knows what is the initial state.
2. Partial knowledge, the operator only knows the initial values of some state variables.
3. No knowledge.

We want to focus on detectability of attacks depending on (1), (2) or (3).

The attack starts at time $t_0 = 0$ and lasts up to a future time T . We define the attack sequence:

$$E(T) = \begin{bmatrix} a(0) \\ a(1) \\ \vdots \\ a(T) \end{bmatrix}$$

It's the vector of attacks from 0 to T .

To study detectability and undetectability, we focus on measurements, and specifically the effect of a on y .

Before this, let's understand how x evolves and how y evolves depending on the attack.

At initial time, we have no impact on the state, as the impact will be seen at x_1 .

We thus have the following:

$$\begin{aligned} & x(0) \text{ initial state} \\ x(1) &= Ax(0) + Ba(0) \\ x(2) &= Ax(1) + Ba(1) = A^2x(0) + ABa(0) + Ba(1) \\ & \vdots \\ x(T+1) &= \dots \end{aligned}$$

Let's put $x(T+1)$ in matrix form.

$$x(T+1) = A^{T+1}x(0) + \underbrace{[A^TB \quad A^{T-1}B \quad \dots \quad AB \quad B]}_{\mathcal{C}_T} \underbrace{\begin{bmatrix} a(0) \\ a(1) \\ \vdots \\ a(T) \end{bmatrix}}_{E(T)}$$

Thus we have:

$$x(T+1) = A^{T+1}x(0) + \mathcal{C}_T \cdot E(T) \longrightarrow \text{Impact}$$

What is now the effect of $E(T)$ on y ?

$$\begin{aligned} y(0) &= Cx(0) + Da(0) \\ y(1) &= Cx(1) + Da(1) = CAx(0) + CBa(0) + Da(1) \\ &\vdots \\ y(T) &= \dots \end{aligned}$$

$y(T)$ in matrix form is:

$$\underbrace{\begin{bmatrix} y(0) \\ \vdots \\ y(T) \end{bmatrix}}_{\mathcal{Y}(T)} = \underbrace{\begin{bmatrix} C \\ CA \\ \vdots \\ CA^T \end{bmatrix}}_{\mathcal{O}(T)} x(0) + \underbrace{\begin{bmatrix} D & 0 & \dots & 0 \\ CB & D & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ CA^{T-1}B & & & D \end{bmatrix}}_{\mathcal{M}_T} \underbrace{\begin{bmatrix} a(0) \\ a(1) \\ \vdots \\ a(T) \end{bmatrix}}_{E(T)}$$

So:

$$\mathcal{Y}(T) = \mathcal{O}_T x(0) + \mathcal{M}_T E(T)$$

The first term is the nominal evolution (due to initial state), and the second is the impact.

Let's now model the knowledge of the initial state. It can be full, partial or zero.

Remember that we distinguish between the initial state of the system, and the actual knowledge the operator has on the initial state. We call this y_Ω .

We have that $y_\Omega = f(x_0)$, a function of the actual initial state. We use a linear function, multiplying the initial state with Ω matrix.

$$y_\Omega = \Omega \cdot x(0)$$

y_Ω has nothing to do with the output, it's the side initial state information about $x(0)$.

The operator does not know $x(0)$, but only y_Ω and Ω .

1. Full knowledge of $x(0)$. What is the condition on Ω matrix? Remember the operator does not directly see $x(0)$. We could choose Ω as an invertible function, so something like I (identity matrix).

Different initial states will be mapped to different initial information.

2. No knowledge of $x(0)$. In this case, $\Omega = 0$ matrix. So $y_\Omega = 0$.
3. Partial knowledge, we have $y_\Omega = \Omega x(0)$ with Ω not full rank. We can distinguish some state variables, but not all.

Suppose we have a mechanical system. In this, the typical state is given by:

$$x = \begin{bmatrix} x_p \\ x_v \end{bmatrix}$$

with x_p being the position, and x_v the velocity.

Suppose we know for sure that the initial velocity is zero, so $x_v(0) = 0$.

We thus have:

$$\begin{bmatrix} 0 & I \end{bmatrix} \begin{bmatrix} x_p(0) \\ x_v(0) \end{bmatrix} = 0$$

Now the detection problem is this: We have the detector, that on side will observe $\mathcal{Y}$, but also y_Ω , and the output will be given by $\Psi(Y(T), y_\Omega)$, and this output can be either 0 or 1 (detection).

Let's see a consistent detector definition.

The detectors is a function that we've seen before, Ψ . taking $\mathcal{Y}(T)$ and y_Ω .

$$\Psi(\mathcal{O}_T x(0) + \mathcal{M}_T E(T), \Omega x(0))$$

Consistency properties want to impose a condition such that there are no false positives.

In nominal conditions, we need to have $\Psi = 0$.

A detector is consistent (no false positives) $\iff \Psi(\mathcal{O}x(0), \Omega x(0)) = 0 \ \forall x(0)$.

We can also see soundness property. A consistent detector Ψ is sound $\iff$:

If $\Psi(\mathcal{Y}(T), \Omega x(0)) = 0$ for Ψ , then it is $\Psi'(\mathcal{Y}(T), \Omega \cdot x(0)) = 0 \ \forall \Psi'$ consistent. The sound detector is the best possible consistent detector.

As for undetectability?

In nominal conditions, given $\mathcal{Y}(T)$ and y_Ω , the detector will output zero.

Under attack, we'll have that $\mathcal{Y}(T) = \mathcal{O}x'(0) + \mathcal{M}_T E(T)$ and $y'_\Omega = \Omega x'(0)$.

When $\mathcal{O}_T x(0) = \mathcal{O}x'(0) + \mathcal{M}_T E(T)$, we have that outputs produced by the attack are compatible with outputs seen in a nominal evolution. Now that the operator has some initial knowledge, we also need to have that $\Omega x(0) = \Omega x'(0)$.

We get:

$$\mathcal{O}(T)(x'(0) - x(0)) + \mathcal{M}_t E(T) = 0$$

$$\Omega(x'(0) - x(0)) = 0$$

Calling $\theta = x'(0) - x(0)$, we have:

$$\mathcal{O}_T \theta + \mathcal{M}_T E(T) = 0$$

and

$$\Omega \theta = 0$$

And so an attack is undetectable $\iff$ these two conditions are met.

In mathematical terms, the two conditions are:

1. $\theta \in \mathcal{V}(\mathcal{E})$, which is the weakly unobservable space. This is the set of all initial states in the system for which we can produce an attack such that the output is zero.
2. $\theta \in \text{Ker}(\Omega)$.

The intersection of this things must not be empty.

What are the particular cases?

1. No information about the initial state, so $\theta = 0$, and we don't have the second condition.
2. Full knowledge: Here $\text{Ker}(\Omega) = 0$, and such $\theta = 0$, and thus $\mathcal{Y}(T) = \mathcal{M}_T E(T) = 0$. Only undetectable attacks are the one having no impact on the output.